MGI-1500

PRODUCT REQUIREMENTS DOCUMENT (PRD)

1500 V Grid-Forming Microgrid Inverter

Industrial Power Conversion / Islandable BESS Microgrids | Rev A2 - Architecture, Atomic Requirements, Interfaces & Verification Baseline

Mission profile. Utility-scale grid-forming inverter for islandable BESS microgrids. The product can establish AC voltage and frequency when the utility grid is unavailable, support black-start and controlled island operation, and reconnect only through an authorized synchronization sequence. The stated sub-cycle hardware overcurrent trip remains independent of grid-control software; its released timing limit is derived from device short-circuit/fault-energy capability rather than assuming that 'sub-cycle' means either a PWM cycle or an AC grid cycle.

Engineering baseline. The power-conversion platform uses a 1500 V DC interface, a three-level NPC inverter with 1700 V IGBT or SiC power modules, VSG control with programmable inertia, liquid-cooled cold plates, IEC 61850 GOOSE communications secured using IEC 62351, NEMA 3R skid integration with an MV transformer, full-power functional verification, HiPot and anti-islanding verification, and lifecycle tracking of semiconductor thermal cycling and filter-capacitor replacement.

Release boundary. The portfolio does not define the released MW rating, AC voltage, nominal frequency, grid-code region, fault-current envelope, efficiency/THD limits, transformer ratio, protection thresholds, coolant limits or environmental severity. Those values remain controlled program parameters. This PRD therefore specifies architecture, ownership, failure behavior and evidence without silently inventing missing numeric limits.

Definition	Controlled product interpretation
Product class	1500 V Grid-Forming Microgrid Inverter
Safety / protection	Grid-forming and grid-support functions are supervised by protection that includes the stated sub-cycle hardware overcurrent trip independent of VSG/grid-control logic. 'Sub-cycle' is retained as a mission-level descriptor; the released trip-time limit shall be allocated from the selected semiconductor/bus fault-energy capability and the complete sensing-to-gate-inhibit delay. Until that allocation is approved, the numerical timing remains TBC.
Lifecycle mission	15-year lifecycle intent with semiconductor thermal-cycle accumulation, DC-link/filter capacitor aging and replacement tracking, cooling-system health evidence, field event logs and controlled second-source/change-impact closure.
Primary architecture	1500 V DC source/BESS interface + precharge/DC link + three-level NPC bridge + AC filter + transformer/switchgear boundary + VSG control + independent fast protection + liquid cooling + IEC 61850/IEC 62351 communications.
Release philosophy	Grid-code, power-rating and installation-dependent values remain visible controlled parameters. Missing qualification, DVP, insulation, protection-coordination or supplier-equivalence evidence blocks release of the affected variant; it is not represented as a fictitious runtime derating mechanism.

PRODUCT GOVERNANCE, CONTROLLED PARAMETERS & REQUIREMENT QUALITY

A grid-forming inverter couples high-energy power hardware, grid-forming control, protection, switchgear, BESS authority and utility interfaces. Product governance separates the released architecture from program-owned grid-code values, site settings and supplier data so a nominal example cannot become an accidental release requirement. Engineering ownership of release governance, traceability and evidence is explicit, including the data, hardware state and external assumptions needed to decide whether the function is valid.

The primary failure concern is an unowned grid-code setting, protection threshold or installation assumption being treated as valid across sites where it has never been verified. Closure uses parameter/TBC register; requirement trace review; release-evidence audit; configuration-status accounting, with the tested revision and active parameter set retained for later change-impact review.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P01-01	Each released product variant shall identify the owner and approved value source for every site-dependent electrical, control, protection, thermal and communication parameter used by the inverter.	parameter/TBC register; requirement trace review
MGI-1500-P01-02	A requirement shall not claim a numeric grid, thermal, fault or environmental limit unless that value is present in the controlled program baseline or has been approved through engineering change control.	requirement trace review; release-evidence audit
MGI-1500-P01-03	Release evidence shall link hardware revision, software build, protection calibration, parameter set, grid-code profile, transformer/switchgear interface and DVP results to one traceable product configuration.	release-evidence audit; configuration-status accounting

Verification, margin and future-change rule

Reassess this page when any of the following changes: grid-code region, BESS variant, transformer, control firmware, protection settings, semiconductor technology or site-interface assumptions. Regression for product governance, controlled parameters and requirement quality shall focus on the assumptions actually changed while preserving traceability to the prior approved evidence.

SYSTEM BOUNDARY, EPC / UTILITY OWNERSHIP & EXPLICIT EXCLUSIONS

The inverter boundary ends at declared DC, AC, cooling, auxiliary-power, control-network and mechanical interfaces. EPC-owned cable runs, MV transformer details, switchgear ratings, protection relays, grounding design and utility interconnection rules remain external unless explicitly allocated to the product. Verification records distinguish product-observable states from site assumptions so ownership remains clear during black start, islanding, faults and service.

The design review explicitly challenges the case of site equipment or protection responsibilities being assumed by both parties or by neither party, producing unsafe energization or incomplete qualification. Objective evidence comes from interface control document; single-line diagram review; responsibility matrix; installation misuse-case review; an evidence gap blocks the affected release claim rather than being disguised as normal runtime behavior.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P02-01	The product shall publish controlled electrical, control, mechanical and cooling interface definitions sufficient for the EPC to complete the site single-line diagram and installation design.	interface control document; single-line diagram review
MGI-1500-P02-02	The inverter shall not energize an AC boundary solely because an external network command is present; local state, protection permissives and installation-authorized interlocks shall also be valid.	single-line diagram review; responsibility matrix
MGI-1500-P02-03	Any installation assumption used to justify insulation, fault current, grounding, thermal or EMC performance shall be traceable to the site interface record and re-reviewed when that assumption changes.	responsibility matrix; installation misuse-case review

Verification, margin and future-change rule

A change to site topology, transformer/switchgear selection, grounding method, BESS interface, auxiliary supply, cooling plant or network ownership requires impact review for product-versus-installation ownership. Requalification scope for system boundary, EPC or utility ownership and explicit exclusions shall be selected from the affected electrical, control, protection, thermal, insulation, network or lifecycle assumptions rather than repeated by default.

OPERATING MODES, STATE AUTHORITY & TRANSITION INTERLOCKS

The operating model includes de-energized service, DC ready, standby, grid-following support where applicable, grid-forming island operation, black start, synchronization, grid-connected operation, controlled shutdown and fault lockout. Mode authority is explicit because the same power stage can be safe in one network condition and hazardous in another. The release argument for mode sequencing and energization authority is built from named interfaces, state authority and the physical limits of the implemented power hardware.

A credible negative-path case must address an invalid transition closing contactors or applying voltage before DC-link, switchgear, synchronization or protection prerequisites are satisfied. The verification basis is state-machine review; transition-table test; interlock fault injection; power-hardware-in-the-loop sequence replay, and results are associated with the hardware, software and site assumptions actually exercised.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P03-01	Every energizing transition shall require a defined set of local hardware, protection, communication and switchgear permissives whose validity is observable by the control state machine.	state-machine review; transition-table test
MGI-1500-P03-02	Loss or contradiction of a required permissive during an energizing sequence shall halt the sequence and drive the inverter to the allocated non-energizing or protected state without relying on a cloud or SCADA response.	transition-table test; interlock fault injection
MGI-1500-P03-03	State-machine DVP shall include interrupted startup, delayed switchgear feedback, stale supervisory commands, DC-source withdrawal and recovery from protection trips.	interlock fault injection; power-hardware-in-the-loop sequence replay

Verification, margin and future-change rule

Future releases shall reopen the mode sequencing and energization authority argument if contactor logic, switchgear feedback, startup sequencing, black-start policy, synchronization logic or supervisory command source changes. The change record for operating modes, state authority and transition interlocks shall identify which verification evidence remains valid and which tests or analyses must be repeated.

1500 V DC INPUT, BESS INTERFACE & DC ISOLATION

The 1500 V DC input is the primary energy boundary between the BESS and inverter. The released design controls polarity, isolation, disconnect status, precharge compatibility, DC-source fault contribution and the voltage/current measurement points used by both control and protection. This page treats 1500 V DC source interface as a controlled engineering capability whose dependencies remain visible across startup, grid events, fault handling and service activity.

Risk analysis focuses on uncontrolled DC energization, reverse connection, ground fault or incorrect source assumptions overstressing the NPC bridge or bypassing safe isolation. Engineering closure combines DC-interface WCCA; insulation review; source-emulator tests; polarity/ground-fault injection; interface FMEA so model, bench and system behavior remain mutually traceable.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P04-01	The inverter shall accept DC energization only through the released BESS/DC-source interface and shall verify the required isolation and disconnect states before enabling the power stage.	DC-interface WCCA; insulation review
MGI-1500-P04-02	DC voltage, current, polarity and insulation information used for energization shall include plausibility diagnostics so a single invalid measurement cannot silently authorize switching.	insulation review; source-emulator tests
MGI-1500-P04-03	The approved DC-input envelope, source impedance assumptions and fault-contribution limits shall be configuration controlled and included in power-stage and protection verification.	polarity/ground-fault injection; interface FMEA

Verification, margin and future-change rule

Configuration control for 1500 V DC source interface includes change triggers covering BESS voltage window, connector/busbar, DC disconnect, insulation monitor, precharge topology or source impedance. Similarity for 1500 V dc input, BESS interface and dc isolation is acceptable only where the unchanged assumptions are demonstrated; otherwise targeted regression is mandatory.

PRECHARGE, DC-LINK ENERGIZATION & DISCHARGE CONTROL

DC-link charging is sequenced to limit inrush and confirm that the inverter bus reaches an expected condition before the main energy path is enabled. The same architecture provides a controlled means to remove hazardous stored energy for shutdown and service. Verification of precharge and DC-link stored-energy control is tied to the exact hardware/software and installation configuration because the surrounding microgrid can materially change the observed behavior.

The failure mode that drives this requirement group is main contactor closure onto an uncharged link, welded precharge hardware, failed discharge or misleading bus-voltage feedback leaving hazardous energy present. The planned evidence - precharge transient test; contactor fault injection; discharge-time verification; bus-voltage plausibility review - is reviewed against the released configuration before the claim is accepted.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P05-01	The inverter shall gate main DC-path closure on measured precharge progress and shall detect timeout, abnormal slope or inconsistent contactor feedback before switching is enabled.	precharge transient test; contactor fault injection
MGI-1500-P05-02	A failed precharge or discharge component shall not be masked by software state progression; the fault shall remain latched or inhibited until the defined recovery conditions are met.	contactor fault injection; discharge-time verification
MGI-1500-P05-03	Precharge and discharge verification shall use beginning- and end-of-life component values, tolerated sensor error and representative BESS source impedance from the controlled program envelope.	discharge-time verification; bus-voltage plausibility review

Verification, margin and future-change rule

Engineering change control shall evaluate DC-link capacitance, precharge resistor, contactor, discharge path, voltage-sense path or source impedance against the evidence on this page. The variant affected by precharge, dc-link energization and discharge control remains unreleased until the required delta analysis and regression are complete.

THREE-LEVEL NPC POWER STAGE & NEUTRAL-POINT BALANCE

The main converter is a three-level neutral-point-clamped architecture. Device command generation, clamping paths and DC-link split-point behavior are coordinated so semiconductor voltage stress and midpoint ripple remain inside the released hardware capability over grid-forming and transient operation. The design therefore gives NPC commutation and neutral-point balance a clear product boundary, with invalid or stale inputs handled separately from incomplete engineering evidence.

The assurance case is incomplete unless it shows how the product contains midpoint drift, illegal switching states or asymmetric loading causing excess device voltage, capacitor stress or distorted output voltage. Verification uses switch-state review; neutral-point simulation; full-power midpoint measurement; fault-state injection; thermal correlation and records any remaining limitation as a controlled open item.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P06-01	PWM generation shall exclude forbidden NPC switch combinations and shall coordinate clamping states with measured DC-link conditions across all enabled operating modes.	switch-state review; neutral-point simulation
MGI-1500-P06-02	Neutral-point imbalance that exceeds the program-approved control range shall invoke the allocated corrective modulation or power limitation before device voltage capability is compromised.	neutral-point simulation; full-power midpoint measurement
MGI-1500-P06-03	The released modulation and midpoint-control strategy shall be validated with representative power factor, unbalance, regenerative operation and dynamic grid-forming events.	fault-state injection; thermal correlation

Verification, margin and future-change rule

The release basis for NPC commutation and neutral-point balance is invalidated by unassessed changes to modulation strategy, clamping device, semiconductor module, DC-link split capacitors, dead time or switching frequency. Preserve the prior three-level npc power stage and neutral-point balance evidence reference and add only the new analysis/test needed to close the delta.

1700 V IGBT / SiC MODULE SELECTION, SOA & DERATING

The product supports controlled 1700 V IGBT or SiC power-module variants. The selected technology is treated as a controlled variant because switching energy, short-circuit withstand, gate requirements, thermal cycling and parasitic sensitivity differ materially between the two implementations. The technical objective is to make power-semiconductor operating margin testable from observable electrical or control states rather than infer correctness from the absence of a trip.

The key concern is not nominal operation but a technology substitution inheriting invalid switching, short-circuit or thermal assumptions and entering production without renewed qualification. Tests and analyses include datasheet/SOA review; loss model; double-pulse testing; junction-temperature correlation; lifetime model to demonstrate the intended response at the relevant boundary conditions.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P07-01	The product configuration shall identify whether the released power stage uses 1700 V IGBT or SiC modules and shall keep the associated gate-drive, SOA and thermal evidence variant specific.	datasheet/SOA review; loss model
MGI-1500-P07-02	Semiconductor operation shall remain within approved voltage, current, switching-energy, junction-temperature and short-circuit boundaries for the declared DC-link and grid transient envelope.	loss model; double-pulse testing
MGI-1500-P07-03	A semiconductor second source or technology migration shall require delta analysis plus targeted electrical, thermal, EMC and protection regression before production equivalence is approved.	junction-temperature correlation; lifetime model

Verification, margin and future-change rule

For later variants, changes involving module vendor, die technology, package, gate resistor, switching frequency, cooling interface or modulation pattern shall be screened against this requirement group before production equivalence is claimed. Any reopened 1700 v IGBT or SiC module selection, soa and derating margin or interface assumption must be closed with objective evidence.

GATE DRIVE, DESATURATION / SHORT-CIRCUIT & SAFE TURN-OFF

Gate drivers translate low-voltage control intent into high-energy switching while providing isolation, UVLO and fast fault protection. The short-circuit path is engineered independently from the slower VSG and supervisory layers because semiconductor survival can depend on microsecond-scale action. Control, protection and installation roles around gate-drive protection and fault turn-off are separated so a supervisory service cannot acquire authority that belongs to local hardware or verified state logic.

Failure analysis examines the possibility of driver UVLO, desaturation failure, common-mode transient or excessive turn-off energy allowing a device fault to propagate through the NPC leg. That question is answered with double-pulse test; short-circuit test; UVLO sweep; CMTI test; fault-turn-off waveform review, including negative-path observation where required to prove the mechanism fails in the intended direction.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P08-01	Each controlled semiconductor position shall have a gate-drive implementation whose UVLO, isolation and fault behavior is compatible with the selected power module and NPC commutation strategy.	double-pulse test; short-circuit test
MGI-1500-P08-02	A detected desaturation or short-circuit event shall suppress the affected switching command through a hardware-capable path that does not depend on completion of the VSG control task.	short-circuit test; UVLO sweep
MGI-1500-P08-03	Turn-off overshoot, fault energy and cross-conduction margin shall be verified on production-representative busbar, module and gate-loop hardware rather than inferred only from schematic simulation.	CMTI test; fault-turn-off waveform review

Verification, margin and future-change rule

Change-impact review is required when driver IC, isolation technology, gate network, Miller clamp, desat blanking, module or busbar parasitics differs from the verified baseline. The review shall determine whether calculation, HIL, bench, full-power or qualification evidence must be repeated for gate-drive protection and fault turn-off.

DC-LINK CAPACITOR BANK, RIPPLE CURRENT & ENERGY CONTAINMENT

The DC-link capacitor bank supports the three-level bridge, absorbs switching ripple and influences fault energy. Split-capacitor voltage balance, ripple heating, discharge behavior and end-of-life capacitance/ESR are included in the electrical and thermal design argument. Engineering ownership of DC-link capacitor electrical and lifetime margin is explicit, including the data, hardware state and external assumptions needed to decide whether the function is valid.

The primary failure concern is ripple-current heating, capacitance loss or imbalance increasing midpoint ripple, device stress or stored-energy risk beyond the verified envelope. Closure uses ripple-current calculation; thermal test; ESR/capacitance aging model; fault-energy review; discharge validation, with the tested revision and active parameter set retained for later change-impact review.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P09-01	The DC-link capacitor bank shall be sized and arranged for the approved ripple-current, voltage-balance and stored-energy conditions associated with the released power stage.	ripple-current calculation; thermal test
MGI-1500-P09-02	Capacitor temperature, imbalance or health indicators used for service decisions shall be based on validated sensing or models and shall not mask an over-stress condition behind nominal bus voltage.	thermal test; ESR/capacitance aging model
MGI-1500-P09-03	Lifecycle evidence shall correlate capacitance and ESR drift to thermal exposure and switching duty and shall define the maintenance trigger used for planned capacitor replacement.	fault-energy review; discharge validation

Verification, margin and future-change rule

Reassess this page when any of the following changes: capacitor family, bank arrangement, cooling path, switching frequency, DC-link ripple spectrum or replacement interval. Regression for dc-link capacitor bank, ripple current and energy containment shall focus on the assumptions actually changed while preserving traceability to the prior approved evidence.

LAMINATED BUSBAR, PARASITICS & COMMUTATION LOOP CONTROL

At 1500 V and multi-megawatt power, stray inductance and capacitance in the DC-link, module and grounding structure materially affect overshoot, EMI and gate-driver stress. The commutation loop is therefore a controlled electrical design item. Parasitic extraction, double-pulse or representative switching tests and layout correlation shall use the released busbar/module stack rather than a nominal interconnect assumption.

The design review explicitly challenges the case of layout or supplier changes increasing commutation inductance enough to defeat the validated voltage margin or EMI behavior. Objective evidence comes from 3D/PEEC extraction or measurement; double-pulse correlation; overshoot measurement; busbar thermal test; an evidence gap blocks the affected release claim rather than being disguised as normal runtime behavior.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P10-01	The released busbar and DC-link geometry shall maintain commutation parasitics within the assumptions used for semiconductor voltage-stress and gate-drive verification.	3D/PEEC extraction or measurement; double-pulse correlation
MGI-1500-P10-02	Mechanical tolerances and assembly torque that influence contact resistance or loop geometry shall be included in manufacturing CTQs where they affect electrical or thermal margin.	double-pulse correlation; overshoot measurement
MGI-1500-P10-03	A busbar, capacitor-terminal or module-footprint change shall trigger targeted overshoot, thermal and EMC regression on representative hardware.	overshoot measurement; busbar thermal test

Verification, margin and future-change rule

A change to busbar geometry, plating, module placement, capacitor terminals, fasteners, insulation stack or grounding geometry requires impact review for high-power interconnect parasitics. Requalification scope for laminated busbar, parasitics and commutation loop control shall be selected from the affected electrical, control, protection, thermal, insulation, network or lifecycle assumptions rather than repeated by default.

AC OUTPUT FILTER, DAMPING & RESONANCE CONTROL

The AC filter limits switching ripple while interacting with grid impedance and grid-forming control. Inductance, capacitance, damping and sensor placement are selected with both harmonic performance and closed-loop stability in view, especially during weak-grid and islanded operation. The release argument for AC filter dynamics and damping is built from named interfaces, state authority and the physical limits of the implemented power hardware.

A credible negative-path case must address an LCL or equivalent resonance coupling with VSG, virtual impedance or external grid impedance and causing oscillation or excessive component stress. The verification basis is impedance model; frequency-response analysis; weak-grid HIL; harmonic measurement; component thermal test, and results are associated with the hardware, software and site assumptions actually exercised.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P11-01	The inverter control and AC-filter design shall be jointly verified for stability over the approved grid-impedance and islanded-load envelope rather than validated as independent subsystems.	impedance model; frequency-response analysis
MGI-1500-P11-02	Filter-component current and voltage stress shall remain within the released component capability during steady state, black start, load steps, faults and reconnection transients.	frequency-response analysis; weak-grid HIL
MGI-1500-P11-03	Changes to filter hardware or control bandwidth shall reopen resonance, harmonic and weak-grid stability evidence before release.	harmonic measurement; component thermal test

Verification, margin and future-change rule

Future releases shall reopen the AC filter dynamics and damping argument if filter L/C values, damping network, switching frequency, control bandwidth, transformer leakage or site grid impedance changes. The change record for ac output filter, damping and resonance control shall identify which verification evidence remains valid and which tests or analyses must be repeated.

MV TRANSFORMER, SWITCHGEAR & AC ISOLATION BOUNDARY

The inverter skid integrates with an MV transformer and switchgear, but exact transformer ratio and protection equipment remain site controlled. The product defines the electrical and signaling assumptions needed for safe energization, isolation, grounding and synchronization at that boundary. This page treats transformer and switchgear interface as a controlled engineering capability whose dependencies remain visible across startup, grid events, fault handling and service activity.

Risk analysis focuses on wrong vector group, grounding assumption, breaker status or transformer impedance invalidating protection, synchronization or harmonic behavior. Engineering closure combines single-line review; transformer data audit; breaker interface test; grounding study interface; energization sequence test so model, bench and system behavior remain mutually traceable.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P12-01	The inverter shall require a controlled transformer/switchgear interface record before site energization and shall not assume a universal MV configuration.	single-line review; transformer data audit
MGI-1500-P12-02	Breaker and isolation status used by the inverter state machine shall be plausibility checked against commanded state and electrical measurements where practical.	transformer data audit; breaker interface test
MGI-1500-P12-03	Transformer or switchgear changes that alter impedance, phase relationship, grounding or protection coordination shall require targeted revalidation of the affected control and protection functions.	grounding study interface; energization sequence test

Verification, margin and future-change rule

Configuration control for transformer and switchgear interface includes change triggers covering transformer design, vector group, grounding, breaker type, CT/VT ratio, switchgear logic or site cable configuration. Similarity for MV transformer, switchgear and ac isolation boundary is acceptable only where the unchanged assumptions are demonstrated; otherwise targeted regression is mandatory.

VOLTAGE / CURRENT SENSING, ISOLATION & MEASUREMENT PLAUSIBILITY

Grid-forming control and protection depend on DC-link, phase voltage, phase current and selected auxiliary measurements. Measurement chains have distinct accuracy, bandwidth, isolation and fault-detection needs; protection-critical sensing is not assumed correct merely because the control loop appears stable. Verification of power measurement integrity is tied to the exact hardware/software and installation configuration because the surrounding microgrid can materially change the observed behavior.

The failure mode that drives this requirement group is sensor offset, saturation, isolation failure or channel swap corrupting VSG control or delaying protection during a high-energy event. The planned evidence - metrology calibration; bandwidth test; sensor fault injection; isolation test; channel-mapping audit - is reviewed against the released configuration before the claim is accepted.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P13-01	Measurement channels used for voltage regulation, current limiting and protection shall have released range, bandwidth, calibration and isolation characteristics appropriate to their allocated functions.	metrology calibration; bandwidth test
MGI-1500-P13-02	The controller shall detect implausible combinations of phase voltage/current, DC power and switch state sufficiently to prevent an invalid channel from silently commanding unsafe power conversion.	bandwidth test; sensor fault injection
MGI-1500-P13-03	Sensor or ADC-path changes shall reopen metrology error budgeting, timing alignment and fault-detection verification for every function that consumes the affected signal.	isolation test; channel-mapping audit

Verification, margin and future-change rule

Engineering change control shall evaluate sensor technology, range, anti-alias filter, ADC, isolation amplifier, CT/VT interface or calibration method against the evidence on this page. The variant affected by voltage or current sensing, isolation and measurement plausibility remains unreleased until the required delta analysis and regression are complete.

CONTROL PLATFORM, SAMPLING, PWM TIMING & COMPUTE RESERVE

The digital control platform executes sampling, transforms, VSG loops, current limiting, modulation, communications and diagnostics with deterministic timing. Compute loading and task priority are controlled so noncritical communications or logging cannot consume the timing reserve needed for power conversion. The design therefore gives real-time control execution a clear product boundary, with invalid or stale inputs handled separately from incomplete engineering evidence.

The assurance case is incomplete unless it shows how the product contains jitter, missed deadlines or resource contention changing phase margin or delaying a protective control action under worst-case software load. Verification uses WCET analysis; scheduler trace; ISR/PWM synchronization test; worst-case CPU load test; watchdog injection and records any remaining limitation as a controlled open item.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P14-01	Sampling, control update and PWM application shall maintain a documented timing relationship across the worst-case released software workload.	WCET analysis; scheduler trace
MGI-1500-P14-02	Lower-criticality diagnostics, network services and logging shall be resource bounded so they cannot starve the control or protection-support tasks required for stable converter operation.	scheduler trace; ISR/PWM synchronization test
MGI-1500-P14-03	Processor, compiler or scheduling changes shall require renewed WCET, jitter and closed-loop timing correlation before production release.	worst-case CPU load test; watchdog injection

Verification, margin and future-change rule

The release basis for real-time control execution is invalidated by unassessed changes to processor, compiler, RTOS, task rates, communication stack, logging load, PWM frequency or control algorithm. Preserve the prior control platform, sampling, pwm timing and compute reserve evidence reference and add only the new analysis/test needed to close the delta.

VIRTUAL SYNCHRONOUS GENERATOR CONTROL & PROGRAMMABLE INERTIA

The VSG function creates grid-forming voltage and frequency behavior rather than following an already established grid reference. Programmable inertia and damping are treated as controlled parameters whose allowable ranges depend on converter current capability, energy source response and microgrid dynamics. The technical objective is to make VSG grid-forming behavior testable from observable electrical or control states rather than infer correctness from the absence of a trip.

The key concern is not nominal operation but an aggressive inertia or damping setting demanding current or energy that the converter/BESS cannot deliver, causing instability or protection cycling. Tests and analyses include control-model review; power-HIL test; load-step response; parameter-sweep stability analysis; BESS interface validation to demonstrate the intended response at the relevant boundary conditions.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P15-01	The VSG controller shall establish AC voltage and frequency in authorized grid-forming modes without requiring a pre-existing utility voltage reference.	control-model review; power-HIL test
MGI-1500-P15-02	Programmable inertia and damping settings shall be constrained to ranges verified against converter current limits, DC-source capability and the approved microgrid dynamic envelope.	power-HIL test; load-step response
MGI-1500-P15-03	VSG parameter-set release shall include model-to-HIL or model-to-power-test correlation for representative islanded loads, disturbances and source limitations.	parameter-sweep stability analysis; BESS interface validation

Verification, margin and future-change rule

For later variants, changes involving VSG algorithm, inertia/damping ranges, BESS power limits, current-limit strategy, filter or microgrid topology shall be screened against this requirement group before production equivalence is claimed. Any reopened virtual synchronous generator control and programmable inertia margin or interface assumption must be closed with objective evidence.

VOLTAGE / FREQUENCY DROOP, VIRTUAL IMPEDANCE & POWER SHARING

Droop and virtual-impedance functions shape real/reactive power sharing, damping and fault response when the inverter operates with other grid-forming sources. Their interaction with feeder impedance and measurement delay is part of the microgrid stability argument. Control, protection and installation roles around droop and virtual-impedance control are separated so a supervisory service cannot acquire authority that belongs to local hardware or verified state logic.

Failure analysis examines the possibility of mismatched droop or virtual impedance causing circulating current, poor reactive sharing or oscillation between paralleled sources. That question is answered with parallel-source HIL; impedance analysis; P/Q step test; circulating-current measurement; parameter sensitivity study, including negative-path observation where required to prove the mechanism fails in the intended direction.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P16-01	The inverter shall apply released active/reactive power droop and virtual-impedance settings only within operating modes for which their interaction has been validated.	parallel-source HIL; impedance analysis
MGI-1500-P16-02	Parallel operation shall bound circulating current and sharing error for the approved combination of source ratings, feeder impedances and communication availability.	impedance analysis; P/Q step test
MGI-1500-P16-03	A change to droop, virtual impedance or site feeder assumptions shall trigger targeted parallel-stability and power-sharing regression.	circulating-current measurement; parameter sensitivity study

Verification, margin and future-change rule

Change-impact review is required when droop coefficients, virtual impedance, feeder impedance, parallel inverter count, BESS limit or transformer configuration differs from the verified baseline. The review shall determine whether calculation, HIL, bench, full-power or qualification evidence must be repeated for droop and virtual-impedance control.

CURRENT LIMITING, PRIORITY LOGIC & CONTROL SATURATION RECOVERY

Grid-forming control must remain physically realizable when loads or faults request more current than the semiconductor stage can supply. Current limiting therefore defines priority between voltage support, real/reactive power and protection while preserving a controlled recovery path after saturation. Engineering ownership of converter current-limit behavior is explicit, including the data, hardware state and external assumptions needed to decide whether the function is valid.

The primary failure concern is integrator windup or incompatible control priorities producing unstable voltage recovery, excessive current or repeated hardware trips. Closure uses current-limit HIL; overload power test; anti-windup review; recovery transient measurement; fault replay, with the tested revision and active parameter set retained for later change-impact review.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P17-01	The control system shall constrain commanded phase current to the approved converter capability before software regulation demands exceed the verified power-stage envelope.	current-limit HIL; overload power test
MGI-1500-P17-02	Entering current limit shall invoke defined control priorities and anti-windup behavior so voltage/frequency regulation can recover without an uncontrolled transient after the overload clears.	overload power test; anti-windup review
MGI-1500-P17-03	Current-limit performance shall be verified for symmetrical and asymmetrical overloads, motor/inrush loads, weak-grid events and the transition into hardware protection.	recovery transient measurement; fault replay

Verification, margin and future-change rule

Reassess this page when any of the following changes: current limits, priority logic, anti-windup method, VSG gains, semiconductor rating or BESS power envelope. Regression for current limiting, priority logic and control saturation recovery shall focus on the assumptions actually changed while preserving traceability to the prior approved evidence.

BLACK-START SEQUENCE & DEAD-BUS ENERGIZATION

Black start is a deliberate grid-forming sequence that energizes a dead AC bus from the BESS. The inverter verifies DC readiness, protection health, switchgear state and load assumptions before ramping voltage/frequency and accepting downstream load. The release sequence identifies which permissives are measured locally and which are supplied by site switchgear/BESS systems so a stale external assumption cannot masquerade as an inverter-ready state.

The design review explicitly challenges the case of energizing a faulted or unexpectedly connected bus, transformer inrush or load pickup exceeding converter capability before protective coordination is established. Objective evidence comes from dead-bus black-start test; transformer/load pickup study; interlock injection; voltage/frequency ramp measurement; an evidence gap blocks the affected release claim rather than being disguised as normal runtime behavior.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P18-01	Black start shall require explicit authorization plus verified DC-source, switchgear and protection conditions before the inverter establishes AC voltage on a de-energized bus.	dead-bus black-start test; transformer/load pickup study
MGI-1500-P18-02	Voltage and frequency build-up shall follow a controlled profile compatible with transformer energization and the approved initial load envelope.	transformer/load pickup study; interlock injection
MGI-1500-P18-03	Black-start DVP shall include failed permissives, unexpected live-bus detection, transformer inrush, staged load pickup and abort/recovery behavior.	interlock injection; voltage/frequency ramp measurement

Verification, margin and future-change rule

A change to black-start sequence, transformer, connected load, BESS reserve, switchgear logic, voltage/frequency ramp profile or protection settings requires impact review for black-start authority and ramp control. Requalification scope for black-start sequence and dead-bus energization shall be selected from the affected electrical, control, protection, thermal, insulation, network or lifecycle assumptions rather than repeated by default.

PLANNED ISLANDING & SEAMLESS MICROGRID FORMATION

During a planned separation from the utility, the control system transfers authority from grid-connected operation to islanded grid formation while coordinating breaker state and local generation/load balance. The sequence is designed to avoid a loss of voltage/frequency reference or conflicting sources. The release argument for intentional islanding transition is built from named interfaces, state authority and the physical limits of the implemented power hardware.

A credible negative-path case must address breaker timing, stale status or insufficient BESS headroom causing frequency excursion, current surge or unintended paralleling during separation. The verification basis is power-HIL islanding test; breaker timing replay; BESS headroom study; frequency/voltage transient measurement, and results are associated with the hardware, software and site assumptions actually exercised.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P19-01	Intentional islanding shall be executed only through an authorized sequence that coordinates switchgear state with the inverter control-mode transition.	power-HIL islanding test; breaker timing replay
MGI-1500-P19-02	The controller shall verify that available BESS/converter capability is consistent with the islanded load envelope before accepting sustained grid-forming responsibility.	breaker timing replay; BESS headroom study
MGI-1500-P19-03	Transition verification shall cover breaker timing dispersion, communication loss, load imbalance and abort paths without relying on nominal event ordering.	BESS headroom study; frequency/voltage transient measurement

Verification, margin and future-change rule

Future releases shall reopen the intentional islanding transition argument if islanding policy, breaker logic, BESS reserve rule, load-shedding coordination, VSG gains or site topology changes. The change record for planned islanding and seamless microgrid formation shall identify which verification evidence remains valid and which tests or analyses must be repeated.

GRID RECONNECTION, SYNCHRONIZATION & BREAKER CLOSING AUTHORITY

Reconnection joins an energized island to the utility and therefore requires validated phase, frequency and voltage alignment plus breaker-authority checks. Synchronization logic is separate from the mechanical breaker action, and stale measurements are not accepted as proof of safe closing conditions. This page treats synchronization and reconnection as a controlled engineering capability whose dependencies remain visible across startup, grid events, fault handling and service activity.

Risk analysis focuses on closing out of phase or on stale/incorrect grid measurements and creating severe current, torque or transformer stress. Engineering closure combines sync-check test; phase-jump injection; measurement-age test; breaker closing-time correlation; reconnection power test so model, bench and system behavior remain mutually traceable.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P20-01	The inverter shall issue a reconnection permissive only when the configured voltage, frequency and phase conditions are satisfied using measurements within their approved freshness and accuracy bounds.	sync-check test; phase-jump injection
MGI-1500-P20-02	Loss of synchronization data, contradictory breaker status or expired measurements shall cancel the closing permissive rather than extrapolating an unsafe connection window.	phase-jump injection; measurement-age test
MGI-1500-P20-03	Synchronization verification shall include worst-case breaker operating time and representative grid phase/frequency disturbances from the approved site envelope.	breaker closing-time correlation; reconnection power test

Verification, margin and future-change rule

Configuration control for synchronization and reconnection includes change triggers covering sync algorithm, VT/measurement chain, breaker closing time, grid-code limits, time source or transformer phase relationship. Similarity for grid reconnection, synchronization and breaker closing authority is acceptable only where the unchanged assumptions are demonstrated; otherwise targeted regression is mandatory.

ANTI-ISLANDING, UNINTENTIONAL ISLAND DETECTION & MODE CONTEXT

Anti-islanding verification is required even though the product also supports intentional island operation. The requirement is therefore mode dependent: an authorized microgrid island is permitted, while an unintentional energized island outside the approved sequence must be detected and cleared according to the site/grid-code concept. Verification of intentional-versus-unintentional island discrimination is tied to the exact hardware/software and installation configuration because the surrounding microgrid can materially change the observed behavior.

The failure mode that drives this requirement group is anti-islanding logic either tripping a legitimate island or failing to clear an unauthorized energized section of network. The planned evidence - mode-dependent anti-islanding test; NDZ study; breaker-state replay; grid-code compliance test plan - is reviewed against the released configuration before the claim is accepted.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P21-01	Anti-islanding behavior shall distinguish authorized grid-forming island modes from conditions in which the inverter is not permitted to sustain an isolated network.	mode-dependent anti-islanding test; NDZ study
MGI-1500-P21-02	Detection and trip logic for an unauthorized island shall not be disabled solely because VSG control can maintain local voltage and frequency.	NDZ study; breaker-state replay
MGI-1500-P21-03	The anti-islanding verification plan shall define the applicable site/grid-code criteria and cover the local generation/load combinations that determine the detection challenge.	breaker-state replay; grid-code compliance test plan

Verification, margin and future-change rule

Engineering change control shall evaluate islanding detection method, grid-code region, breaker topology, local load/generation mix, synchronization logic or measurement filters against the evidence on this page. The variant affected by anti-islanding, unintentional island detection and mode context remains unreleased until the required delta analysis and regression are complete.

PARALLEL GRID-FORMING SOURCES & MULTI-INVERTER COORDINATION

Utility-scale microgrids may operate multiple grid-forming inverters or other sources in parallel. The product therefore defines how local droop, synchronization, communications and current limits cooperate without creating a single point of centralized control dependency. The design therefore gives parallel source coordination a clear product boundary, with invalid or stale inputs handled separately from incomplete engineering evidence.

The assurance case is incomplete unless it shows how the product contains two sources establishing incompatible references, fighting through circulating current or losing stable sharing when a communication link fails. Verification uses multi-inverter HIL; parallel power test; communication-drop test; sharing-error analysis; faulted-unit removal test and records any remaining limitation as a controlled open item.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P22-01	Each inverter shall retain sufficient local grid-forming authority to remain stable within the approved parallel architecture when nonessential supervisory communications are lost.	multi-inverter HIL; parallel power test
MGI-1500-P22-02	Parallel synchronization and power sharing shall avoid uncontrolled circulating current during unit addition, removal and step changes in load or BESS capability.	parallel power test; communication-drop test
MGI-1500-P22-03	The released parallel-operation matrix shall identify verified inverter counts, rating combinations and coordination settings; untested combinations remain unreleased.	sharing-error analysis; faulted-unit removal test

Verification, margin and future-change rule

The release basis for parallel source coordination is invalidated by unassessed changes to parallel unit count, rating mix, droop settings, communication architecture, feeder impedance or master/peer policy. Preserve the prior parallel grid-forming sources and multi-inverter coordination evidence reference and add only the new analysis/test needed to close the delta.

BESS / EMS POWER AUTHORITY, SOC RESERVE & COMMAND ARBITRATION

The inverter draws energy from an external BESS and receives dispatch or mode requests from an EMS/SCADA layer. Grid-forming control can request rapid power, but BESS limits and reserve policy remain authoritative constraints on sustainable operation. The technical objective is to make BESS and supervisory command arbitration testable from observable electrical or control states rather than infer correctness from the absence of a trip.

The key concern is not nominal operation but a stale dispatch, unavailable SOC reserve or conflicting limit causing the VSG to demand power the BESS cannot safely provide. Tests and analyses include interface contract review; stale-command test; BESS limit replay; power-authority arbitration HIL to demonstrate the intended response at the relevant boundary conditions.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P23-01	The inverter shall consume BESS charge/discharge limits and supervisory commands through interfaces with defined validity, freshness and authority semantics.	interface contract review; stale-command test
MGI-1500-P23-02	Local current, thermal and protection limits shall take precedence over an external dispatch request that would exceed the released converter or BESS interface capability.	stale-command test; BESS limit replay
MGI-1500-P23-03	Loss of valid EMS data shall lead to a defined locally safe operating policy rather than continued use of an unbounded stale power command.	BESS limit replay; power-authority arbitration HIL

Verification, margin and future-change rule

For later variants, changes involving BESS limits, SOC reserve policy, EMS protocol, dispatch semantics, timeout rules or grid-forming reserve strategy shall be screened against this requirement group before production equivalence is claimed. Any reopened BESS or ems power authority, soc reserve and command arbitration margin or interface assumption must be closed with objective evidence.

GRID-CODE / SITE PROFILE & SETPOINT ENVELOPE MANAGEMENT

Frequency, voltage, reactive-power, ride-through and anti-islanding requirements vary by utility and jurisdiction. This PRD does not invent a universal grid code; instead, it requires an identified site profile whose approved settings bound the relevant control and protection functions. Control, protection and installation roles around site-specific grid compliance parameters are separated so a supervisory service cannot acquire authority that belongs to local hardware or verified state logic.

Failure analysis examines the possibility of a parameter file intended for one jurisdiction or transformer configuration being deployed to another site without compatibility evidence. That question is answered with profile checksum/version audit; settings review; site acceptance test; configuration-lock test; change-control records, including negative-path observation where required to prove the mechanism fails in the intended direction.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P24-01	The active grid/site profile shall be uniquely identified and cryptographically or procedurally protected against unauthorized modification according to the project cybersecurity concept.	profile checksum/version audit; settings review
MGI-1500-P24-02	Control and protection functions that consume site parameters shall validate profile compatibility with the hardware/software variant before enabling grid-connected operation.	settings review; site acceptance test
MGI-1500-P24-03	A new jurisdiction or utility profile shall remain unreleased until the applicable simulation, HIL, protection and site-acceptance evidence has been completed.	configuration-lock test; change-control records

Verification, margin and future-change rule

Change-impact review is required when utility requirements, jurisdiction, transformer ratio, nominal frequency/voltage, protection limits or firmware interpretation of the profile differs from the verified baseline. The review shall determine whether calculation, HIL, bench, full-power or qualification evidence must be repeated for site-specific grid compliance parameters.

PROTECTION COORDINATION WITH MV RELAYS, BREAKERS & FUSES

The inverter contains fast local protection but operates inside a wider site protection scheme that may include MV relays, breakers, fuses and BESS protection. Selectivity and timing are coordinated so a local semiconductor event is cleared quickly without relying on a slower external relay, while system faults do not create unintended protection races. Engineering ownership of multi-layer protection coordination is explicit, including the data, hardware state and external assumptions needed to decide whether the function is valid.

The primary failure concern is overlapping or missing protection zones causing either unnecessary plant trips or delayed clearing of a high-energy fault. Closure uses protection coordination study; time-current curves; trip-injection test; breaker timing correlation; fault matrix review, with the tested revision and active parameter set retained for later change-impact review.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P25-01	The protection concept shall document which faults are cleared locally by the inverter and which rely on coordinated site equipment, including timing and isolation ownership.	protection coordination study; time-current curves
MGI-1500-P25-02	Local semiconductor protection shall not wait for an IEC 61850 message or MV relay operation when the allocated device-survival response requires faster action.	time-current curves; trip-injection test
MGI-1500-P25-03	Site protection changes that alter clearing time or fault contribution shall trigger targeted coordination review and fault-injection regression.	breaker timing correlation; fault matrix review

Verification, margin and future-change rule

Reassess this page when any of the following changes: relay settings, breaker clearing time, fuse, transformer, BESS protection, inverter current limit or local trip threshold. Regression for protection coordination with MV relays, breakers and fuses shall focus on the assumptions actually changed while preserving traceability to the prior approved evidence.

SUB-CYCLE HARDWARE OVERCURRENT TRIP INDEPENDENCE

The independent hardware overcurrent path has its own sensor, threshold, propagation, latch and switch-disable chain outside the VSG/grid-control software. The portfolio term 'sub-cycle' is not used as an unquantified acceptance criterion: the project shall allocate a maximum sensing-to-gate-inhibit delay from the selected semiconductor short-circuit withstand, busbar/fault-energy limits and gate-driver behavior, with the exact numerical limit controlled in the protection parameter register.

Protection verification challenges software overload, corrupted control measurements and unstable VSG behavior while measuring the independent trip path from fault stimulus to gate inhibition. Hardware fault injection, propagation-delay measurement, threshold WCCA, independence review and repeated short-circuit/fault-energy validation shall demonstrate margin to the selected device and power-stage limits. Missing timing or withstand evidence blocks release of the affected power-stage variant.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P26-01	The inverter shall implement a hardware-capable overcurrent trip whose actuation does not require execution of the VSG, current-control or supervisory software tasks.	hardware fault-injection test; propagation-delay measurement
MGI-1500-P26-02	The complete trip chain from fault sensing through gate inhibition shall satisfy the program-approved sub-cycle response allocation under worst-case component, temperature and supply tolerance.	propagation-delay measurement; independence review
MGI-1500-P26-03	Fast-trip verification shall demonstrate independence from software hang, network loss and control-loop saturation and shall be repeated for protection-path or semiconductor changes.	threshold WCCA; repeated short-circuit validation

Verification, margin and future-change rule

A change to current sensor, comparator/protection IC, threshold network, driver interface, latch logic, semiconductor or gate-drive delay requires impact review of the complete fault-energy path. Regression shall repeat the timing, WCCA and short-circuit/fault-energy evidence affected by the change; the generic word 'sub-cycle' shall not substitute for the released numerical delay requirement.

DC GROUND FAULT, INSULATION MONITORING & FAULT CONTAINMENT

The 1500 V DC system requires a defined response to insulation degradation and ground faults consistent with the BESS and site grounding concept. The inverter owns the interpretation of its local measurements and the permissive logic allocated to it, while the exact grounding topology remains installation controlled. The release argument for DC insulation and ground-fault behavior is built from named interfaces, state authority and the physical limits of the implemented power hardware.

A credible negative-path case must address an undetected ground fault reducing insulation margin or combining with a second fault to create hazardous current through the power stage or enclosure. The verification basis is insulation-monitor injection; ground-fault test; grounding concept review; isolation measurement correlation, and results are associated with the hardware, software and site assumptions actually exercised.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P27-01	The inverter shall monitor or receive the DC insulation status required by the site safety concept before and during energization.	insulation-monitor injection; ground-fault test
MGI-1500-P27-02	A confirmed insulation fault shall invoke the allocated inhibit, trip or controlled shutdown response without being overridden by a grid-support request.	ground-fault test; grounding concept review
MGI-1500-P27-03	Changes that alter common-mode capacitance or grounding shall reopen insulation-monitor interpretation, leakage-current and fault-containment evidence.	grounding concept review; isolation measurement correlation

Verification, margin and future-change rule

Future releases shall reopen the DC insulation and ground-fault behavior argument if grounding method, insulation monitor, filter capacitance to ground, BESS isolation scheme, transformer or cable architecture changes. The change record for dc ground fault, insulation monitoring and fault containment shall identify which verification evidence remains valid and which tests or analyses must be repeated.

AC OVER / UNDER VOLTAGE, FREQUENCY & ABNORMAL-GRID SUPERVISION

Grid-connected and islanded modes use different authority for voltage and frequency, but both require supervision of values outside the approved operating envelope. Protection and ride-through thresholds are sourced from the controlled site profile rather than embedded as universal numbers. This page treats abnormal AC condition supervision as a controlled engineering capability whose dependencies remain visible across startup, grid events, fault handling and service activity.

Risk analysis focuses on control continuing to energize through a condition that exceeds the site protection envelope or tripping unnecessarily during an explicitly required ride-through event. Engineering closure combines threshold sweep; ride-through HIL; sensor fault injection; event-log review; profile/version audit so model, bench and system behavior remain mutually traceable.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P28-01	The inverter shall supervise AC voltage and frequency against the active site profile and shall distinguish normal, ride-through, islanded and trip regions according to the released mode logic.	threshold sweep; ride-through HIL
MGI-1500-P28-02	Measurement invalidity or profile mismatch shall prevent use of the affected protection decision until a defined safe fallback or shutdown condition is reached.	ride-through HIL; sensor fault injection
MGI-1500-P28-03	Threshold and timing verification shall include sensor tolerance, filter delay and control-mode transitions relevant to the selected grid-code profile.	event-log review; profile/version audit

Verification, margin and future-change rule

Configuration control for abnormal AC condition supervision includes change triggers covering site grid profile, nominal frequency, voltage scaling, ride-through curve, measurement filters or protection allocation. Similarity for ac over or under voltage, frequency and abnormal-grid supervision is acceptable only where the unchanged assumptions are demonstrated; otherwise targeted regression is mandatory.

FAULT RIDE-THROUGH, VOLTAGE SUPPORT & POST-FAULT RECOVERY

Where the site profile requires ride-through, the inverter may remain connected during defined grid disturbances while respecting current and thermal limits. Grid-forming voltage support is constrained by physical converter capability and protection priorities rather than by an unconditional mandate to hold nominal voltage. Verification of ride-through and recovery control is tied to the exact hardware/software and installation configuration because the surrounding microgrid can materially change the observed behavior.

The failure mode that drives this requirement group is support logic driving excessive current or failing to recover stable VSG operation after a voltage sag, phase jump or fault clearing event. The planned evidence - sag/swell HIL; fault ride-through power test; current-limit trace; post-fault stability analysis; protection coordination review - is reviewed against the released configuration before the claim is accepted.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P29-01	The inverter shall execute only those fault ride-through behaviors that are enabled by the approved site profile and verified against current, DC-source and thermal capability.	sag/swell HIL; fault ride-through power test
MGI-1500-P29-02	During a ride-through event, hardware protection remains authoritative even when continued grid support is requested by control software or an external utility command.	fault ride-through power test; current-limit trace
MGI-1500-P29-03	Post-fault recovery shall avoid uncontrolled voltage/frequency overshoot and shall be verified for representative clearing times and weak-grid conditions.	post-fault stability analysis; protection coordination review

Verification, margin and future-change rule

Engineering change control shall evaluate ride-through profile, current priority, VSG gains, grid impedance, semiconductor limits or transformer configuration against the evidence on this page. The variant affected by fault ride-through, voltage support and post-fault recovery remains unreleased until the required delta analysis and regression are complete.

UNBALANCED LOADS, NEGATIVE SEQUENCE & PHASE-SPECIFIC LIMITING

Islanded microgrids and faults can produce phase unbalance and negative-sequence current. The controller manages those conditions within converter and transformer capability while preserving a stable voltage reference for the healthy phases. The design therefore gives unbalanced three-phase operation a clear product boundary, with invalid or stale inputs handled separately from incomplete engineering evidence.

The assurance case is incomplete unless it shows how the product contains negative-sequence heating, neutral-point stress or a single overloaded phase exceeding semiconductor current while total three-phase power appears acceptable. Verification uses unbalanced-load power test; negative-sequence analysis; phase-current limiting test; thermal correlation and records any remaining limitation as a controlled open item.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P30-01	Phase-current protection and limiting shall operate on phase-specific conditions so an overloaded phase cannot be hidden by average three-phase power.	unbalanced-load power test; negative-sequence analysis
MGI-1500-P30-02	The released controller shall define how positive/negative-sequence components are treated in islanded and grid-connected modes consistent with transformer and semiconductor thermal limits.	negative-sequence analysis; phase-current limiting test
MGI-1500-P30-03	Unbalanced-load DVP shall include single-phase load steps, asymmetrical faults and recovery without loss of protection selectivity.	phase-current limiting test; thermal correlation

Verification, margin and future-change rule

The release basis for unbalanced three-phase operation is invalidated by unassessed changes to phase-current limits, transformer grounding, modulation, filter, VSG sequence control or load envelope. Preserve the prior unbalanced loads, negative sequence and phase-specific limiting evidence reference and add only the new analysis/test needed to close the delta.

LIQUID-COOLED THERMAL ARCHITECTURE & MULTI-MEGAWATT LOSS BUDGET

The power stage uses liquid-cooled cold plates to remove semiconductor and associated power-component losses at multi-megawatt scale. Thermal ownership includes coolant inlet conditions, flow distribution, contact interfaces, sensors and the loss model used to predict junction and component temperatures. The technical objective is to make system thermal path and cooling capacity testable from observable electrical or control states rather than infer correctness from the absence of a trip.

The key concern is not nominal operation but uneven coolant distribution, underestimated switching loss or degraded thermal interfaces creating a local hot spot not visible from a single bulk coolant measurement. Tests and analyses include loss breakdown; CFD/thermal model; full-power heat run; sensor correlation; flow-imbalance test to demonstrate the intended response at the relevant boundary conditions.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P31-01	The thermal design shall correlate electrical loss for each major power component to the released coolant and ambient boundary conditions used for product qualification.	loss breakdown; CFD/thermal model
MGI-1500-P31-02	Temperature sensing and derating logic shall detect or conservatively bound local thermal conditions that can diverge from average cold-plate temperature.	CFD/thermal model; full-power heat run
MGI-1500-P31-03	A cooling or loss-model change shall reopen full-power thermal correlation and lifetime calculations for the affected hardware variant.	sensor correlation; flow-imbalance test

Verification, margin and future-change rule

For later variants, changes involving power rating, semiconductor, switching frequency, cold plate, coolant, pump/flow architecture, TIM or enclosure airflow shall be screened against this requirement group before production equivalence is claimed. Any reopened liquid-cooled thermal architecture and multi-megawatt loss budget margin or interface assumption must be closed with objective evidence.

COLD PLATES, COOLANT FLOW, LEAK DETECTION & SERVICE BOUNDARY

Cold plates are part of the power-stage reliability chain and must maintain electrical isolation, hydraulic integrity and thermal contact over service life. The inverter also defines what coolant flow/temperature information is required from any external cooling plant. Control, protection and installation roles around cooling hardware integrity and diagnostics are separated so a supervisory service cannot acquire authority that belongs to local hardware or verified state logic.

Failure analysis examines the possibility of flow blockage, trapped air, leakage or cold-plate degradation causing rapid thermal rise or compromising insulation without timely detection. That question is answered with pressure/leak test; flow-loss injection; thermal runaway test; coolant compatibility aging; service procedure review, including negative-path observation where required to prove the mechanism fails in the intended direction.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P32-01	The inverter shall inhibit or limit power when required coolant conditions are invalid according to the released thermal protection concept.	pressure/leak test; flow-loss injection
MGI-1500-P32-02	Cooling-system faults shall be distinguished from sensor faults where practical so service diagnostics do not authorize continued full power on an unverified thermal path.	flow-loss injection; thermal runaway test
MGI-1500-P32-03	Cold-plate and coolant changes shall require compatibility, leakage, pressure, thermal and electrical-isolation evidence before production release.	coolant compatibility aging; service procedure review

Verification, margin and future-change rule

Change-impact review is required when cold plate, seals, coolant chemistry, hose/fitting, external pump, flow sensor, leak sensor or maintenance interval differs from the verified baseline. The review shall determine whether calculation, HIL, bench, full-power or qualification evidence must be repeated for cooling hardware integrity and diagnostics.

SEMICONDUCTOR THERMAL CYCLING, DAMAGE ACCUMULATION & HEALTH ESTIMATION

IGBT thermal-cycling accumulation tracking is a lifecycle requirement; equivalent technology-specific evidence is required for a SiC variant. Junction-temperature swing, dwell, mean temperature and mission profile are converted into a controlled damage indicator linked to maintenance and design review. Engineering ownership of power-module lifetime consumption is explicit, including the data, hardware state and external assumptions needed to decide whether the function is valid.

The primary failure concern is bond-wire, solder or package fatigue progressing faster than the lifecycle model predicts because actual field thermal cycles differ from the assumed mission profile. Closure uses mission-profile analysis; junction-temperature reconstruction; power cycling correlation; damage-model validation; field trend review, with the tested revision and active parameter set retained for later change-impact review.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P33-01	The product shall accumulate a power-module thermal-cycle health metric using a model correlated to the selected semiconductor/package technology and available temperature information.	mission-profile analysis; junction-temperature reconstruction
MGI-1500-P33-02	Lifecycle diagnostics shall retain sufficient field history to distinguish high cumulative damage from a short transient temperature event.	junction-temperature reconstruction; power cycling correlation
MGI-1500-P33-03	Changes to module technology or thermal mission profile shall reopen lifetime-model correlation and maintenance threshold approval.	damage-model validation; field trend review

Verification, margin and future-change rule

Reassess this page when any of the following changes: power module, cooling, switching strategy, mission profile, thermal sensor model, damage algorithm or service-life target. Regression for semiconductor thermal cycling, damage accumulation and health shall focus on the assumptions actually changed while preserving traceability to the prior approved evidence.

FILTER / DC-LINK CAPACITOR AGING & REPLACEMENT TRACKING

The 15-year lifecycle baseline includes replacement tracking for filter capacitors. Capacitor health depends on ripple current, voltage, temperature and storage history, so the maintenance strategy uses those stressors rather than a calendar interval alone unless deliberately defined by the program. Service thresholds are tied to measured/estimated capacitor stress and the released capacitor family.

The design review explicitly challenges the case of ESR growth or capacitance loss increasing filter resonance, ripple heating or DC-link stress before scheduled maintenance recognizes the degradation. Objective evidence comes from accelerated aging; ripple/temperature model; capacitance/ESR inspection; service threshold review; field trend analysis; an evidence gap blocks the affected release claim rather than being disguised as normal runtime behavior.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P34-01	The inverter lifecycle record shall track the capacitor stresses or approved proxy metrics used to justify inspection or replacement of DC-link and AC-filter capacitor populations.	accelerated aging; ripple/temperature model
MGI-1500-P34-02	A capacitor health indication shall not be used to extend service life beyond the validated model envelope without engineering approval and supporting evidence.	ripple/temperature model; capacitance/ESR inspection
MGI-1500-P34-03	Replacement parts shall preserve the electrical, thermal and harmonic assumptions of the released design or trigger targeted requalification.	service threshold review; field trend analysis

Verification, margin and future-change rule

A change to capacitor supplier, dielectric, bank sizing, cooling, harmonic spectrum, service interval or health algorithm requires impact review for capacitor lifecycle and service planning. Requalification scope for filter or dc-link capacitor aging and replacement tracking shall be selected from the affected electrical, control, protection, thermal, insulation, network or lifecycle assumptions rather than repeated by default.

AUXILIARY POWER, CONTROL SUPPLIES & RIDE-THROUGH OF LV SERVICES

Control processors, gate drivers, protection, communications, relays and cooling instrumentation depend on low-voltage auxiliary supplies that may behave differently during black start or grid outages. The LV architecture therefore preserves essential control/protection long enough to reach a defined state when a source changes or collapses. The release argument for auxiliary-power continuity and sequencing is built from named interfaces, state authority and the physical limits of the implemented power hardware.

A credible negative-path case must address a control supply brownout issuing partial gate commands, losing trip latches or corrupting nonvolatile state during a high-energy transition. The verification basis is power-tree review; brownout test; source-transfer test; reset-sequence capture; NVM interruption test, and results are associated with the hardware, software and site assumptions actually exercised.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P35-01	Essential protection and gate-disable functions shall retain defined behavior throughout auxiliary-supply undervoltage and reset transitions.	power-tree review; brownout test
MGI-1500-P35-02	The control platform shall detect and log brownout/reset causes needed to distinguish a power-source problem from an inverter control fault.	brownout test; source-transfer test
MGI-1500-P35-03	Auxiliary-power changes shall be verified with black-start, grid-loss and controlled-shutdown sequences using production-representative loads.	reset-sequence capture; NVM interruption test

Verification, margin and future-change rule

Future releases shall reopen the auxiliary-power continuity and sequencing argument if auxiliary source, DC/DC, hold-up capacitance, reset supervisor, gate supply, cooling controls or sequencing firmware changes. The change record for auxiliary power, control supplies and ride-through of lv services shall identify which verification evidence remains valid and which tests or analyses must be repeated.

IEC 61850 NETWORK ARCHITECTURE, DATA OWNERSHIP & SERVICE BOUNDARY

IEC 61850 communications connect the inverter to station automation and protection infrastructure. The product defines which logical data and commands it publishes, subscribes to or treats as authoritative; network services remain subordinate to local electrical protection. This page treats station-network communication ownership as a controlled engineering capability whose dependencies remain visible across startup, grid events, fault handling and service activity.

Risk analysis focuses on an ambiguous GOOSE/control mapping giving external equipment unintended authority over energization or protection-critical behavior. Engineering closure combines IEC 61850 data-model review; SCL/configuration audit; command-authority test; network-loss test so model, bench and system behavior remain mutually traceable.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P36-01	The released IEC 61850 configuration shall identify each subscribed and published item, its semantic owner and whether it is advisory, permissive, operational or protection related.	IEC 61850 data-model review; SCL/configuration audit
MGI-1500-P36-02	No network command shall bypass local hardware protection or energization interlocks allocated to the inverter.	SCL/configuration audit; command-authority test
MGI-1500-P36-03	Changes to SCL/data mapping or station architecture shall trigger interface regression and command-authority review before deployment.	command-authority test; network-loss test

Verification, margin and future-change rule

Configuration control for station-network communication ownership includes change triggers covering data model, SCL file, switch architecture, command mapping, station controller or firmware communication stack. Similarity for IEC 61850 network architecture, data ownership and service boundary is acceptable only where the unchanged assumptions are demonstrated; otherwise targeted regression is mandatory.

GOOSE TIMING, FRESHNESS, DUPLICATION & LOSS-OF-NETWORK BEHAVIOR

GOOSE messaging can carry time-sensitive station events, but deterministic network behavior is verified rather than assumed. Sequence number, freshness, duplicate handling and loss-of-network behavior are defined for each function that consumes GOOSE information. Verification of GOOSE event integrity and timing is tied to the exact hardware/software and installation configuration because the surrounding microgrid can materially change the observed behavior.

The failure mode that drives this requirement group is stale, duplicated or delayed event data being interpreted as current switchgear or protection state during an energization transition. The planned evidence - network impairment test; latency measurement; duplicate/replay injection; switch failover test; application trace - is reviewed against the released configuration before the claim is accepted.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P37-01	A GOOSE subscriber shall validate the message state, sequence/freshness information and configured publisher identity before using the event in control logic.	network impairment test; latency measurement
MGI-1500-P37-02	Loss or expiry of a GOOSE input shall drive the function-specific fallback defined in the interface contract rather than preserving an indefinite last-known permissive.	latency measurement; duplicate/replay injection
MGI-1500-P37-03	GOOSE timing verification shall include network loading, switch failover and representative fault/event bursts for the released station architecture.	switch failover test; application trace

Verification, margin and future-change rule

Engineering change control shall evaluate network switch, VLAN/QoS, GOOSE mapping, retransmission profile, firmware stack or site topology against the evidence on this page. The variant affected by GOOSE timing, freshness, duplication and loss-of-network behavior remains unreleased until the required delta analysis and regression are complete.

IEC 62351 SECURITY, CERTIFICATES & TRUST-ANCHOR MANAGEMENT

IEC 61850 GOOSE communications are secured using the project-selected IEC 62351 mechanisms. Security therefore covers trusted identities, certificate/key lifecycle, protected configuration and operational recovery; cryptography is not treated as a substitute for electrical safety or local protection. The design therefore gives IEC 62351 trust and credential lifecycle a clear product boundary, with invalid or stale inputs handled separately from incomplete engineering evidence.

The assurance case is incomplete unless it shows how the product contains expired, compromised or mismatched credentials either admitting unauthorized station traffic or blocking legitimate recovery during a grid event. Verification uses certificate lifecycle test; trust-anchor audit; revoked/expired credential test; provisioning record review; secure configuration test and records any remaining limitation as a controlled open item.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P38-01	The inverter shall authenticate and protect applicable IEC 61850 communications using the IEC 62351 mechanisms selected by the project security architecture.	certificate lifecycle test; trust-anchor audit
MGI-1500-P38-02	Credential expiry, revocation and replacement shall have defined operational behavior that preserves local protection and prevents unauthorized control access.	trust-anchor audit; revoked/expired credential test
MGI-1500-P38-03	Trust-anchor or credential-store changes shall require security regression and provisioning traceability before a new software/hardware image is released.	provisioning record review; secure configuration test

Verification, margin and future-change rule

The release basis for IEC 62351 trust and credential lifecycle is invalidated by unassessed changes to PKI, certificate profile, key store, trust anchor, crypto library, station domain or maintenance process. Preserve the prior IEC 62351 security, certificates and trust-anchor management evidence reference and add only the new analysis/test needed to close the delta.

SECURE BOOT, SOFTWARE UPDATE, ROLLBACK & CALIBRATION AUTHORITY

Control software and site calibration can directly affect grid-forming behavior. Boot authenticity, update authorization, rollback and parameter compatibility are therefore treated as product-security and release-integrity functions rather than convenience features. The technical objective is to make software and configuration integrity testable from observable electrical or control states rather than infer correctness from the absence of a trip.

The key concern is not nominal operation but an unauthorized or incompatible image changing protection interaction, VSG behavior or grid-code settings without the release evidence associated with that combination. Tests and analyses include secure-boot negative test; signed-update test; rollback/recovery test; calibration compatibility audit; image genealogy review to demonstrate the intended response at the relevant boundary conditions.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P39-01	The inverter shall verify the authenticity and compatibility of executable software before enabling power conversion.	secure-boot negative test; signed-update test
MGI-1500-P39-02	An interrupted or failed update shall recover to an approved image or remain non-energizing; it shall not boot an unverified partial image.	signed-update test; rollback/recovery test
MGI-1500-P39-03	Software/calibration release records shall preserve the exact protection, VSG and site-profile versions used for verification and field deployment.	calibration compatibility audit; image genealogy review

Verification, margin and future-change rule

For later variants, changes involving bootloader, signing keys, firmware image, calibration schema, update service, storage device or recovery partition shall be screened against this requirement group before production equivalence is claimed. Any reopened secure boot, software update, rollback and calibration authority margin or interface assumption must be closed with objective evidence.

TIME SYNCHRONIZATION, EVENT LOGGING & DISTURBANCE RECORDING

Accurate event reconstruction is essential for utility incidents, protection analysis and fleet reliability. The inverter timestamps mode changes, trips, key measurements and network events with declared clock quality so post-event data is not presented as more precise than the available time source supports. Control, protection and installation roles around time quality and forensic event evidence are separated so a supervisory service cannot acquire authority that belongs to local hardware or verified state logic.

Failure analysis examines the possibility of clock jumps, stale time or log overflow obscuring the sequence between inverter action, breaker operation and external grid events. That question is answered with time-loss test; clock-step test; event-sequence correlation; buffer-overflow test; disturbance-record review, including negative-path observation where required to prove the mechanism fails in the intended direction.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P40-01	Event records shall include timestamp-quality or synchronization-status information sufficient to interpret their sequence during grid and protection investigations.	time-loss test; clock-step test
MGI-1500-P40-02	Loss of external time synchronization shall not disable local control or protection; the inverter shall continue with defined local timing while flagging reduced timestamp confidence.	clock-step test; event-sequence correlation
MGI-1500-P40-03	Logging and disturbance-record capacity shall be verified for the event burst expected during representative protection and reconnection sequences.	buffer-overflow test; disturbance-record review

Verification, margin and future-change rule

Change-impact review is required when time source, synchronization protocol, logging rate, storage size, event schema or station network architecture differs from the verified baseline. The review shall determine whether calculation, HIL, bench, full-power or qualification evidence must be repeated for time quality and forensic event evidence.

NEMA 3R SKID, OUTDOOR ENVIRONMENT & MV INTEGRATION

The product is integrated as NEMA 3R outdoor-rated equipment with the MV transformer and associated skid hardware. Rain, solar heating, contamination, condensation, service access and cable entry are handled as installation-aware design conditions rather than a generic indoor electronics envelope. Engineering ownership of outdoor skid environmental integrity is explicit, including the data, hardware state and external assumptions needed to decide whether the function is valid.

The primary failure concern is water entry, condensation or field-service changes degrading insulation, cooling, corrosion resistance or high-voltage clearances. Closure uses NEMA enclosure inspection; rain/ingress test; thermal solar-load review; condensation assessment; service-access audit, with the tested revision and active parameter set retained for later change-impact review.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P41-01	The enclosure and skid integration shall maintain the released NEMA 3R environmental function for all approved cable-entry, service-door and ventilation configurations.	NEMA enclosure inspection; rain/ingress test
MGI-1500-P41-02	Outdoor thermal and moisture conditions shall be included in the insulation and cooling qualification used to authorize full-power operation.	rain/ingress test; thermal solar-load review
MGI-1500-P41-03	Mechanical or enclosure changes that alter airflow, ingress paths, creepage/clearance or cable routing shall require targeted environmental and electrical requalification.	condensation assessment; service-access audit

Verification, margin and future-change rule

Reassess this page when any of the following changes: enclosure, vent/louver, door/seal, cable gland, skid arrangement, transformer proximity or outdoor location. Regression for NEMA 3R skid, outdoor environment and MV integration shall focus on the assumptions actually changed while preserving traceability to the prior approved evidence.

GROUNDING, BONDING, LIGHTNING / SURGE & COMMON-MODE CONTROL

High-power outdoor equipment must manage protective bonding, common-mode current and surge energy across DC, AC, auxiliary and communication boundaries. The inverter interfaces with the site grounding and lightning-protection system without assuming external protection will absorb every internal transient. Grounding and surge verification uses the released skid bonding, cable entry and protection-device arrangement together with the site coordination assumptions.

The design review explicitly challenges the case of poor bonding or surge-current paths elevating touch potential, damaging insulation or coupling disruptive common-mode energy into control electronics. Objective evidence comes from ground-bond test; surge test; lightning-current path review; common-mode measurement; site grounding interface audit; an evidence gap blocks the affected release claim rather than being disguised as normal runtime behavior.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P42-01	Protective bonding and shield-termination points shall be defined in the released mechanical/electrical interface and verified for the intended fault and surge-current paths.	ground-bond test; surge test
MGI-1500-P42-02	Surge-protection components shall be coordinated with insulation capability and expected site exposure so clamping does not create an unverified secondary stress elsewhere in the inverter.	surge test; lightning-current path review
MGI-1500-P42-03	Grounding or surge-path changes shall reopen EMC, insulation, touch-potential and protection-coordination evidence as applicable.	common-mode measurement; site grounding interface audit

Verification, margin and future-change rule

A change to bonding strap, cable shield termination, SPD, enclosure, transformer grounding, filter Y-capacitance or site earthing requires impact review for grounding and surge robustness. Requalification scope for grounding, bonding, lightning or surge and common-mode control shall be selected from the affected electrical, control, protection, thermal, insulation, network or lifecycle assumptions rather than repeated by default.

EMC EMISSIONS, IMMUNITY & HIGH-POWER SWITCHING COEXISTENCE

The NPC bridge, gate drives, DC bus and station network coexist within a high dV/dt and dI/dt environment. EMC design therefore addresses conducted/radiated emissions, immunity of sensing/control, common-mode paths and the possibility that noise corrupts protection or communication rather than only causing a regulatory failure. The release argument for EMC robustness at full switching power is built from named interfaces, state authority and the physical limits of the implemented power hardware.

A credible negative-path case must address switching noise causing false trips, sensor corruption, network errors or excessive external emissions after a layout or semiconductor change. The verification basis is pre-compliance scan; full-power EMC test; immunity test; probe correlation; communication-error monitoring, and results are associated with the hardware, software and site assumptions actually exercised.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P43-01	The inverter shall meet the program-approved EMC limits for the released installation while preserving correct control and protection behavior during immunity exposure.	pre-compliance scan; full-power EMC test
MGI-1500-P43-02	EMC testing shall monitor functional effects such as false trips, ADC corruption, watchdog resets and communication faults in addition to measuring emissions.	full-power EMC test; immunity test
MGI-1500-P43-03	Changes that materially affect switching edge rate or common-mode paths shall trigger targeted EMC regression before production equivalence is approved.	probe correlation; communication-error monitoring

Verification, margin and future-change rule

Future releases shall reopen the EMC robustness at full switching power argument if semiconductor, gate speed, busbar, filter, cable shielding, grounding, enclosure or network hardware changes. The change record for emc emissions, immunity and high-power switching coexistence shall identify which verification evidence remains valid and which tests or analyses must be repeated.

CREEPAGE, CLEARANCE, HiPot & PARTIAL-DISCHARGE / INSULATION MARGIN

The 1500 V DC architecture and MV-connected skid require controlled insulation distances and dielectric qualification across semiconductor, busbar, cooling, sensing and enclosure interfaces. Exact insulation category and test voltages remain program controlled; the evidence must nevertheless correspond to the released physical stack. This page treats high-voltage insulation system as a controlled engineering capability whose dependencies remain visible across startup, grid events, fault handling and service activity.

Risk analysis focuses on contamination, tolerance stack-up or cooling/mechanical changes reducing dielectric margin despite a nominal schematic remaining unchanged. Engineering closure combines creepage/clearance audit; HiPot test; insulation resistance; partial-discharge evaluation where applicable; tolerance review so model, bench and system behavior remain mutually traceable.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P44-01	The released hardware shall maintain the program-approved creepage, clearance and dielectric withstand requirements across manufacturing tolerance and environmental exposure.	creepage/clearance audit; HiPot test
MGI-1500-P44-02	HiPot and insulation tests shall be performed at defined manufacturing/EOL points without relying on a single final test to screen defects that can damage hardware earlier in the process.	HiPot test; insulation resistance
MGI-1500-P44-03	Insulation-system material or geometry changes shall require documented delta analysis and targeted dielectric requalification.	partial-discharge evaluation where applicable; tolerance review

Verification, margin and future-change rule

Configuration control for high-voltage insulation system includes change triggers covering busbar insulation, PCB stack-up, cold plate, sensor isolation, connector, enclosure, altitude/environmental category or dielectric material. Similarity for creepage, clearance, HiPot and partial-discharge or insulation margin is acceptable only where the unchanged assumptions are demonstrated; otherwise targeted regression is mandatory.

MECHANICAL STRUCTURE, BUS SUPPORT, VIBRATION & TRANSPORT LOADS

The skid and internal structures restrain heavy magnetic components, busbars, capacitor banks and cooling hardware during transport, installation and service. Mechanical motion is evaluated for its electrical consequences, including loosening, insulation abrasion and thermal-interface degradation. Verification of mechanical integrity of high-power assemblies is tied to the exact hardware/software and installation configuration because the surrounding microgrid can materially change the observed behavior.

The failure mode that drives this requirement group is transport shock or vibration changing bus spacing, fastener preload or cooling contact enough to create a latent electrical or thermal defect. The planned evidence - structural analysis; vibration/shock test; torque audit; post-test HiPot; post-test thermal/electrical inspection - is reviewed against the released configuration before the claim is accepted.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P45-01	High-energy components and bus structures shall retain required alignment, clearances and connection integrity through the approved transport and installation load cases.	structural analysis; vibration/shock test
MGI-1500-P45-02	Post-mechanical-qualification inspections shall include electrical and thermal checks where movement could affect insulation or contact resistance.	vibration/shock test; torque audit
MGI-1500-P45-03	Structural or mounting changes shall reopen the affected vibration, torque, clearance and thermal-interface evidence rather than being approved on drawing similarity alone.	post-test HiPot; post-test thermal/electrical inspection

Verification, margin and future-change rule

Engineering change control shall evaluate skid frame, mounting, bus support, transformer interface, capacitor bracket, fastener or transport method against the evidence on this page. The variant affected by mechanical structure, bus support, vibration and transport loads remains unreleased until the required delta analysis and regression are complete.

MANUFACTURING CTQs, ASSEMBLY TORQUE & COOLING CLEANLINESS

Production controls focus on characteristics that materially affect high-power electrical, thermal and insulation performance. Examples include power-terminal torque, busbar spacing, TIM application, coolant cleanliness, seal integrity and sensor/calibration identity. The design therefore gives manufacturing process capability a clear product boundary, with invalid or stale inputs handled separately from incomplete engineering evidence.

The assurance case is incomplete unless it shows how the product contains a process shift creating high contact resistance, coolant contamination or reduced insulation margin that passes basic functional test but fails at full power. Verification uses PFMEA/control plan; torque traceability; cleanliness limits; process capability; inspection MSA; golden-unit correlation and records any remaining limitation as a controlled open item.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P46-01	Manufacturing shall identify and control the CTQs that influence power-terminal resistance, insulation, cooling performance and calibrated measurement accuracy.	PFMEA/control plan; torque traceability
MGI-1500-P46-02	Process records shall provide traceability from inverter serial number to critical component lots, torque/assembly data, firmware/configuration and EOL results as required by the release plan.	torque traceability; cleanliness limits
MGI-1500-P46-03	Insufficient process-capability or measurement-system evidence shall block production release of the affected process; it shall not be represented as a runtime inverter response.	inspection MSA; golden-unit correlation

Verification, margin and future-change rule

The release basis for manufacturing process capability is invalidated by unassessed changes to assembly process, tool, supplier, TIM dispense, coolant fill, cleaning chemistry, fixture or acceptance criteria. Preserve the prior manufacturing ctqs, assembly torque and cooling cleanliness evidence reference and add only the new analysis/test needed to close the delta.

EOL FUNCTIONAL TEST, FULL-POWER VERIFICATION & HiPot

Production acceptance includes full-power functional verification, HiPot, and anti-islanding sequence verification. Production EOL therefore combines electrical safety screening with calibrated functional checks that demonstrate correct phase mapping, sensing, switching, cooling and configuration before a unit is released. The technical objective is to make end-of-line electrical and functional acceptance testable from observable electrical or control states rather than infer correctness from the absence of a trip.

The key concern is not nominal operation but a wiring, assembly or calibration defect escaping low-power test and appearing only during high-current operation or site commissioning. Tests and analyses include EOL station qualification; full-power run; HiPot records; measurement uncertainty; golden-unit correlation; serial genealogy to demonstrate the intended response at the relevant boundary conditions.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P47-01	Each production unit shall complete the released EOL sequence for identity, insulation, sensing, switching, cooling and functional behavior before shipment.	EOL station qualification; full-power run
MGI-1500-P47-02	Where the manufacturing concept claims full-power EOL coverage, the applied power and duration shall be sufficient to expose the targeted assembly and thermal defects defined by the PFMEA.	full-power run; HiPot records
MGI-1500-P47-03	A failed or incomplete EOL record shall prevent unit release until dispositioned through the controlled nonconformance process.	golden-unit correlation; serial genealogy

Verification, margin and future-change rule

For later variants, changes involving EOL fixture, test sequence, measurement instrument, acceptance limit, software image, calibration or power-stage variant shall be screened against this requirement group before production equivalence is claimed. Any reopened eol functional test, full-power verification and HiPot margin or interface assumption must be closed with objective evidence.

EOL ANTI-ISLANDING, SYNCHRONIZATION & PROTECTION-INJECTION TEST

Dynamic sequence tests at EOL or factory acceptance confirm that the configured control/protection image matches the hardware and site profile. Anti-islanding, synchronization and protection-injection checks target command authority and timing without assuming that a successful steady-state power run proves these behaviors. Control, protection and installation roles around dynamic EOL / FAT protection sequence evidence are separated so a supervisory service cannot acquire authority that belongs to local hardware or verified state logic.

Failure analysis examines the possibility of wrong configuration or I/O mapping allowing correct nominal power conversion while failing to trip, synchronize or recognize switchgear states correctly. That question is answered with sequence test; simulated breaker I/O; current/voltage injection; GOOSE simulation; trip timing capture, including negative-path observation where required to prove the mechanism fails in the intended direction.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P48-01	Factory acceptance shall exercise the released anti-islanding and synchronization logic with representative switchgear status and electrical inputs for the configured project variant.	sequence test; simulated breaker I/O
MGI-1500-P48-02	Protection injection shall confirm the expected local trip path and recorded event identity without depending on live utility faults to validate the mechanism.	simulated breaker I/O; current/voltage injection
MGI-1500-P48-03	Configuration or I/O changes after FAT shall be subject to documented regression or site-acceptance testing before energization.	GOOSE simulation; trip timing capture

Verification, margin and future-change rule

Change-impact review is required when I/O map, site profile, protection calibration, firmware, station network mapping, FAT fixture or breaker interface differs from the verified baseline. The review shall determine whether calculation, HIL, bench, full-power or qualification evidence must be repeated for dynamic EOL / FAT protection sequence evidence.

EFFICIENCY, HARMONICS, POWER QUALITY & STEADY-STATE DVP

Product performance includes conversion efficiency, harmonic behavior, power factor/reactive capability and stable operation over the program-defined DC, AC and thermal envelope. Exact acceptance values remain project controlled, but the DVP must measure them with uncertainty and configuration traceability. Engineering ownership of steady-state electrical performance is explicit, including the data, hardware state and external assumptions needed to decide whether the function is valid.

The primary failure concern is a control or component change improving one operating point while degrading harmonics, thermal margin or efficiency elsewhere in the required envelope. Closure uses calorimetric/electrical loss test; harmonic analyzer; P/Q map; thermal soak; measurement uncertainty analysis, with the tested revision and active parameter set retained for later change-impact review.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P49-01	Steady-state DVP shall map efficiency, current/voltage quality and thermal performance over the released operating envelope rather than relying on a single nominal point.	calorimetric/electrical loss test; harmonic analyzer
MGI-1500-P49-02	Power-quality measurements shall identify the test bandwidth, instrument uncertainty and site/filter configuration used for the acceptance claim.	harmonic analyzer; P/Q map
MGI-1500-P49-03	Performance changes outside the tested interpolation range shall be treated as open evidence until validated by analysis or targeted test.	thermal soak; measurement uncertainty analysis

Verification, margin and future-change rule

Reassess this page when any of the following changes: modulation, switching frequency, power module, filter, transformer, control gains, coolant conditions or performance limits. Regression for efficiency, harmonics, power quality and steady-state DVP shall focus on the assumptions actually changed while preserving traceability to the prior approved evidence.

GRID-FORMING DVP: BLACK START, ISLAND, WEAK GRID & PARALLEL OPERATION

A dedicated grid-forming DVP demonstrates the behaviors that distinguish this product from a conventional grid-following inverter. Scenarios combine black start, intentional islanding, weak-grid interaction, parallel-source operation, large load steps and recovery from current limiting. Each result records the exercised BESS, transformer, switchgear, grid-strength and control configuration so site assumptions remain traceable to the measured response.

The design review explicitly challenges the case of individual loops passing unit tests while the integrated VSG, current limits, filter and switchgear sequence become unstable in realistic microgrid events. Objective evidence comes from power-HIL scenario matrix; full-power microgrid test; weak-grid emulator; parallel source testing; event-record review; an evidence gap blocks the affected release claim rather than being disguised as normal runtime behavior.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P50-01	The product-level DVP shall include end-to-end scenarios that exercise grid formation and transition behavior with the same software, protection calibration and hardware variant intended for release.	power-HIL scenario matrix; full-power microgrid test
MGI-1500-P50-02	Acceptance shall evaluate voltage/frequency stability, current-limit behavior, protection interaction and recovery rather than only whether the inverter remained energized.	full-power microgrid test; weak-grid emulator
MGI-1500-P50-03	Open grid-forming DVP evidence shall keep the affected operating mode or product variant unreleased until the defined verification gate is closed.	parallel source testing; event-record review

Verification, margin and future-change rule

A change to VSG, droop, current limit, filter, BESS source, transformer, switchgear logic or site topology requires impact review for end-to-end grid-forming performance. Requalification scope for grid-forming DVP: black start, island, weak grid and parallel shall be selected from the affected electrical, control, protection, thermal, insulation, network or lifecycle assumptions rather than repeated by default.

FAULT-INJECTION, PROTECTION INDEPENDENCE & NEGATIVE-PATH DVP

Nominal power tests cannot establish protection integrity. Fault-injection testing therefore targets sensor faults, stuck commands, delayed breaker feedback, communication loss, software overload, gate-driver faults and fast overcurrent paths to verify that the intended independent mechanisms actually dominate. The release argument for negative-path and independence validation is built from named interfaces, state authority and the physical limits of the implemented power hardware.

A credible negative-path case must address a hidden common dependency allowing a single processor, power rail or measurement fault to defeat both control and protection claims. The verification basis is fault-injection matrix; independence analysis; software hang test; sensor substitution; protection timing capture; common-cause review, and results are associated with the hardware, software and site assumptions actually exercised.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P51-01	The verification plan shall inject representative single faults into control, sensing, communication and protection paths and confirm the allocated containment or shutdown response.	fault-injection matrix; independence analysis
MGI-1500-P51-02	Claims of protection independence shall be supported by physical and logical dependency analysis, not only by separate software functions running on the same resources.	independence analysis; software hang test
MGI-1500-P51-03	Any change that introduces a new shared resource between control and protection shall reopen common-cause and independence verification.	protection timing capture; common-cause review

Verification, margin and future-change rule

Future releases shall reopen the negative-path and independence validation argument if protection architecture, control processor, power supply, sensing, gate drive, communication stack or diagnostic coverage changes. The change record for fault-injection, protection independence and negative-path DVP shall identify which verification evidence remains valid and which tests or analyses must be repeated.

ENVIRONMENTAL / SEQUENTIAL QUALIFICATION RELEASE GATE

Outdoor environmental qualification is a release decision tied to the exact hardware and installation variant. Temperature, moisture, vibration, contamination and electrical stress may interact, so sequential testing or justified correlation is used where combined aging can expose defects missed by isolated tests. This page treats environmental qualification evidence as a controlled engineering capability whose dependencies remain visible across startup, grid events, fault handling and service activity.

Risk analysis focuses on an enclosure, seal, insulation or cooling variant entering production on similarity assumptions without evidence that combined environmental stress preserves the safety and performance margins. Engineering closure combines qualification matrix; sequential stress test; post-stress HiPot; post-stress full-power check; configuration-status audit so model, bench and system behavior remain mutually traceable.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P52-01	Environmental qualification shall be associated with a defined hardware, material and installation configuration and shall record any allowed family/variant similarity rationale.	qualification matrix; sequential stress test
MGI-1500-P52-02	Post-stress verification shall include the electrical, thermal and insulation checks needed to reveal degradation relevant to the stressed design features.	sequential stress test; post-stress HiPot
MGI-1500-P52-03	Missing environmental qualification evidence shall block release of the affected hardware or installation variant rather than trigger a fictitious runtime derating state.	post-stress full-power check; configuration-status audit

Verification, margin and future-change rule

Configuration control for environmental qualification evidence includes change triggers covering enclosure, seal, cooling, dielectric material, outdoor rating, mounting, supplier or environmental category. Similarity for environmental or sequential qualification release gate is acceptable only where the unchanged assumptions are demonstrated; otherwise targeted regression is mandatory.

15-YEAR LIFECYCLE, MAINTENANCE & FIELD HEALTH MANAGEMENT

The lifecycle plan combines semiconductor thermal-cycle accumulation, capacitor replacement tracking, cooling-system service, event history and obsolescence management over the 15-year mission. Field health metrics support maintenance decisions but do not replace component qualification or approved inspection intervals without evidence. Verification of long-term product health and maintenance is tied to the exact hardware/software and installation configuration because the surrounding microgrid can materially change the observed behavior.

The failure mode that drives this requirement group is aging mechanisms accumulating outside the model envelope until a latent power, cooling or insulation failure appears without a planned service response. The planned evidence - mission-profile model; field-health trend; maintenance plan; accelerated aging correlation; service record audit - is reviewed against the released configuration before the claim is accepted.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P53-01	The product lifecycle record shall retain the health and stress information needed to support the approved maintenance strategy for power modules, capacitors and cooling hardware.	mission-profile model; field-health trend
MGI-1500-P53-02	Service thresholds derived from lifetime models shall remain linked to the component technology and mission profile used to validate those models.	field-health trend; maintenance plan
MGI-1500-P53-03	A lifecycle model or maintenance-policy change shall require engineering approval and targeted correlation before it is applied to deployed fleets.	accelerated aging correlation; service record audit

Verification, margin and future-change rule

Engineering change control shall evaluate mission profile, semiconductor, capacitor, cooling hardware, maintenance interval, health algorithm or service-life target against the evidence on this page. The variant affected by 15-year lifecycle, maintenance and field health management remains unreleased until the required delta analysis and regression are complete.

SUPPLIER / SECOND-SOURCE EQUIVALENCE & OBSOLESCENCE CONTROL

Long-life utility products require controlled component substitutions, but equivalence is established by affected physics rather than form-fit-function declarations alone. Power modules, capacitors, gate drivers, sensors, cooling parts, insulation materials and network components each carry different regression obligations. The design therefore gives supplier change and production equivalence a clear product boundary, with invalid or stale inputs handled separately from incomplete engineering evidence.

The assurance case is incomplete unless it shows how the product contains a second source altering switching loss, dielectric margin, thermal impedance, communication timing or lifetime while retaining the same nominal specification. Verification uses supplier delta analysis; WCCA update; targeted regression; qualification evidence; change-control board approval and records any remaining limitation as a controlled open item.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P54-01	Second-source approval shall identify which electrical, thermal, insulation, control, EMC, cybersecurity and lifecycle assumptions are affected by the proposed change.	supplier delta analysis; WCCA update
MGI-1500-P54-02	Production equivalence shall not be granted until the required delta analysis and targeted regression evidence are closed for the exact replacement configuration.	WCCA update; targeted regression
MGI-1500-P54-03	Obsolescence planning shall preserve traceability from fielded serial numbers to affected component populations and approved replacement paths.	qualification evidence; change-control board approval

Verification, margin and future-change rule

The release basis for supplier change and production equivalence is invalidated by unassessed changes to supplier, component process, material, firmware dependency, lifecycle status, qualification status or manufacturing site. Preserve the prior supplier or second-source equivalence and obsolescence control evidence reference and add only the new analysis/test needed to close the delta.

FINAL RELEASE CLOSURE, OPEN-ITEM CONTROL & FUTURE-CHANGE GOVERNANCE

Final release joins architecture, protection, grid-forming DVP, insulation, thermal, manufacturing, cybersecurity and lifecycle evidence into one controlled product baseline. Open items are explicit and owned; unresolved grid-code or site values remain TBC rather than being hidden behind nominal examples. The technical objective is to make production-release evidence and change governance testable from observable electrical or control states rather than infer correctness from the absence of a trip.

The key concern is not nominal operation but future software, component or site-profile changes invalidating a key grid-forming or protection assumption without triggering the necessary regression. Tests and analyses include release checklist; open-item audit; configuration baseline; change-impact matrix; release-board approval to demonstrate the intended response at the relevant boundary conditions.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
MGI-1500-P55-01	Production release shall require closure or formally approved disposition of the architecture, protection, DVP, insulation, thermal, manufacturing, cybersecurity and lifecycle evidence allocated to the product variant.	release checklist; open-item audit
MGI-1500-P55-02	Open TBC items shall identify owner, affected requirement and release impact; an unresolved item shall not be silently converted into a nominal product requirement.	open-item audit; configuration baseline
MGI-1500-P55-03	Post-release engineering changes shall use impact analysis to select the necessary regression scope while preserving traceability to the prior approved evidence baseline.	change-impact matrix; release-board approval

Verification, margin and future-change rule

For later variants, changes involving any power-stage, control, protection, thermal, insulation, network, site-profile, manufacturing or lifecycle assumption that supports an approved requirement shall be screened against this requirement group before production equivalence is claimed. Any reopened final release closure, open-item control and future-change governance margin or interface assumption must be closed with objective evidence.